Assessment 4

DDD 2: Introduction to Programming and Basic Data Analysis

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Setup

```
rm(list = ls())
```

Load Required Libraries

```
# Library loading
library(tibble)
library(dplyr)
library(rlang)
library(lobstr)
library(rio)
library(ggplot2)
library(purrr)
library(labelled)
library(stargazer)
library(scales)
library(janitor)
library(flextable)
```

IHDS Data exercise

```
library(tidyverse)
library(rio)
HH_DATA <- import("../data/ddd2/RAW_DATA/ICPSR_36151/HH_DATA/HH_DATA.dta")

IHDS<- subset(HH_DATA, select= c(ID18C, ID13, URBAN2011))
view(IHDS)
```

Assessment Environment

```
#All assessment answers and variables stored here in this environment
assessment_3 <- new.env()
```

Question 1: Data Cleaning and Visualization

Step 1

```
assessment_3$raw_data   <-   import("../data/ddd2/RAW_DATA/ICPSR_36151/HH_DATA/
HH_DATA.dta")
```

Step 2

```
# STEP 2: Select required variables
library(tidyverse)
assessment_3$df <- assessment_3$raw_data |>
  select(ID11, ID13, STATEID, INCOMEPC) |>
  rename(
    region = ID11,
    caste = ID13,
    state_id = STATEID,
    income_pc = INCOMEPC)|> filter(!is.na(state_id))
```

Step 3 : Create new variable for geographical zones

```
assessment_3$df <- assessment_3$df |>
  mutate(
    geo_zone = case_when(
      state_id %in% c(2, 3, 6, 7, 12, 5, 22) ~ "North",
      state_id %in% c(8, 24) ~ "West",
      state_id %in% c(18, 23) ~ "Central",
      state_id %in% c(29, 33, 36, 28, 32) ~ "South",
      state_id %in% c(19, 10, 20, 21, 4) ~ "East",
      TRUE ~ NA_character_))
```

Step 4 : Clean data by removing rows with missing values in key variables

```
assessment_3$df_clean <- assessment_3$df |>
  filter(!is.na(geo_zone), !is.na(caste), !is.na(income_pc))
```

Step 5 : Contingency table - Caste vs Zones don

```
assessment_3$df_clean |>
  mutate(across(c(caste, geo_zone), as.factor)) |>
  tabyl(geo_zone, caste) |>
  adorn_percentages("row") |>
  adorn_pct_formatting() |>
  janitor::adorn_ns() |>
  flextable()
```

geo_zone	1	2	3	4	5	6
Central	5.8% (236)	20.7% (849)	38.9% (1,593)	18.1% (741)	15.8% (646)	0.9% (35)
East	5.4% (377)	26.5% (1,848)	33.4% (2,327)	24.4% (1,699)	9.7% (676)	0.7% (50)
North	8.0% (628)	27.9% (2,181)	27.4% (2,142)	29.2% (2,285)	7.4% (577)	0.2% (17)
South	1.8% (174)	12.8% (1,232)	55.9% (5,366)	20.2% (1,940)	5.4% (521)	3.9% (370)
West	4.7% (216)	20.6% (949)	47.4% (2,181)	17.4% (800)	8.7% (398)	1.2% (54)

#Step 6 : Summary statistics of income by zone

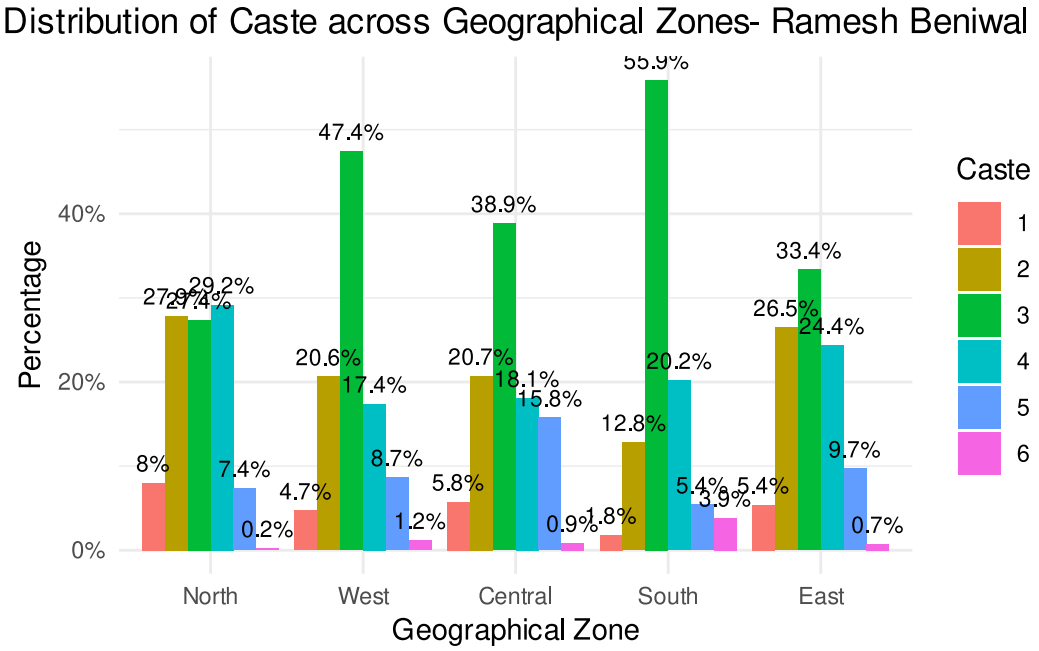
```
assessment_3$df_clean |>
  group_by(geo_zone) |>
  summarise(
    n = n(),
    mean_income = mean(income_pc, na.rm = TRUE),
    min_income = min(income_pc, na.rm = TRUE),
    max_income = max(income_pc, na.rm = TRUE))
```

```
# A tibble: 5 × 5
  geo_zone     n mean_income min_income max_income
  <chr>    <int>       <dbl>      <dbl>      <dbl>
1 Central   4100      21229.    -19800     770333.
2 East      6977      23900.    -22726.   2792455
3 North     7830      39457.    -45366.   4161000
4 South     9603      31070.   -289008.   2698850
5 West      4598      28324.   -259260    1650000
```

PLOT 1: Caste distribution across zones

```
assessment_3$df_clean |>
  filter(!is.na(caste), !is.na(geo_zone)) |>
  count(geo_zone, caste) |>
  group_by(geo_zone) |>   # <-- important fix
  mutate(prop = n / sum(n),top_lab = paste0(round(prop * 100, 1), "%")) |>
  ungroup() |>
  ggplot(aes(
```

```
 x = factor(geo_zone, levels = c("North", "West", "Central", "South", "East")),
  y = prop,
  fill = factor(caste))) +
geom_col(position = position_dodge(width = 0.9)) +
geom_text(
  aes(label = top_lab),
  position = position_dodge(width = 0.9),
  vjust = -0.5,
  size = 3) +
scale_y_continuous(labels = scales::percent) +
labs(
  x = "Geographical Zone",
  y = "Percentage",
  title = "Distribution of Caste across Geographical Zones- Ramesh Beniwal",
  fill = "Caste") +
theme_minimal() +
theme(
  plot.title = element_text(hjust = 0.5),
  legend.position = "right")
```

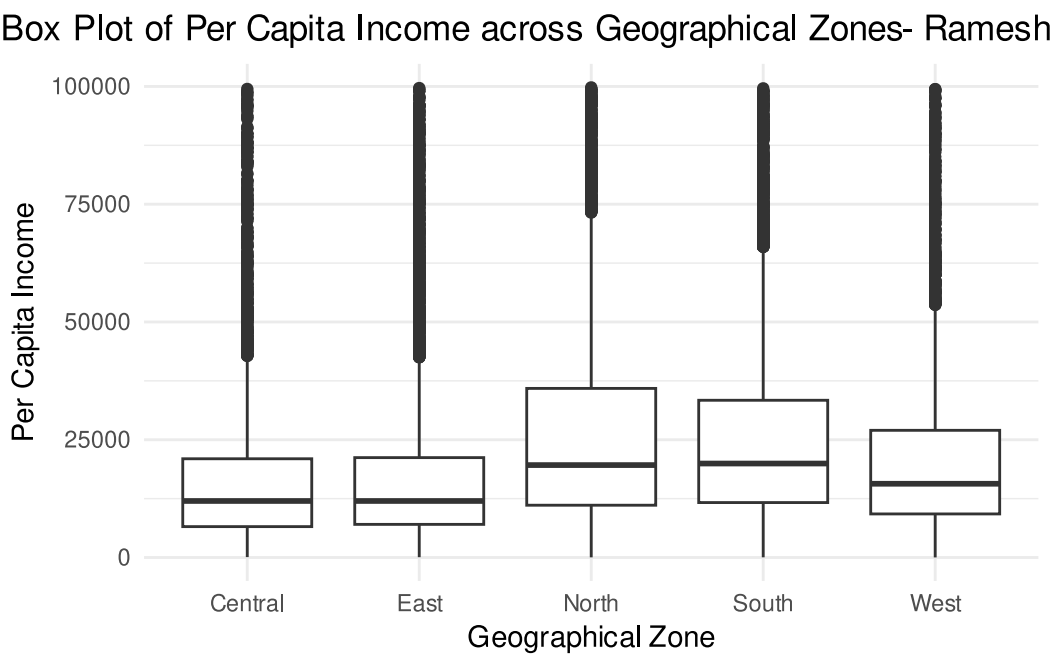


Interpretatiion of the above plot: The chart shows that Caste 3 has the highest percentage in all regions, with the strongest presence in the South. Caste 2 and Caste 4 appear in moderate proportions across every zone. Caste 1 and Caste 6 have very low representation everywhere. The North region looks more balanced compared to others, with several castes having similar values. In contrast, the West and South are more dominated by one major caste group. The East and Central regions show a mixed pattern without extreme dominance. Overall, caste distribution is uneven and changes from one geographical zone to another.

---

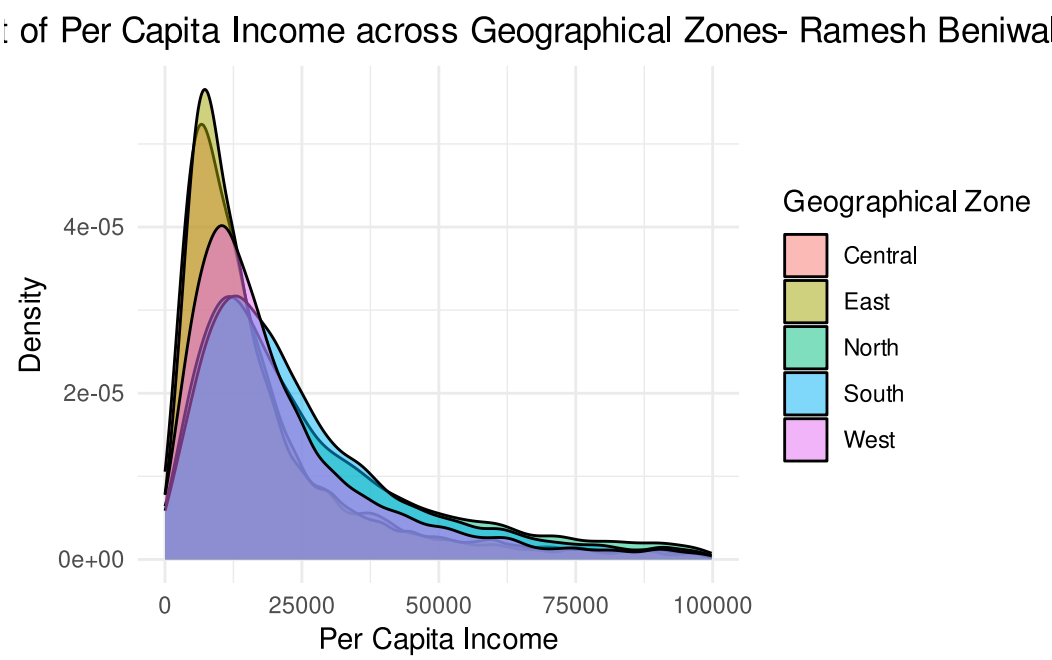
plot box to check outliers

```
assessment_3$df_clean |>
  filter(!is.na(income_pc), !is.na(geo_zone)) |>
  filter(income_pc > 0, income_pc < 100000) |>
  ggplot(aes(x = geo_zone, y = income_pc)) +
  geom_boxplot() +
  labs(
    x = "Geographical Zone",
    y = "Per Capita Income",
     title = "Box Plot of Per Capita Income across Geographical Zones- Ramesh
Beniwal") +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5),
    legend.position = "right")
```



PLOT 2 : Density plot the Distribution of per capita income across zone and remove outliers from income

```
assessment_3$df_clean |>
  filter(!is.na(income_pc), !is.na(geo_zone)) |>
  filter(income_pc > 0, income_pc < 100000) |>
  ggplot(aes(x = income_pc, fill = geo_zone)) +
  geom_density(alpha = 0.5) +
  labs(
    x = "Per Capita Income",
    y = "Density",
      title = "Density Plot of Per Capita Income across Geographical Zones-
Ramesh Beniwal",
    fill = "Geographical Zone") +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5),
    legend.position = "right")
```



Interpretation of the above plot: The density plot shows that most people in all regions fall in the lower income range, as the curves peak at lower values. The East and Central regions have slightly higher peaks, indicating more people concentrated at lower incomes. The North and South regions show a wider spread, suggesting more variation in income levels. The West also follows a similar pattern but with a slightly smoother distribution. All regions have a long right tail, meaning a smaller number of people earn much higher incomes. Overall, income distribution is positively skewed across all geographical zones.

Final Check

```
ls(assessment_3)
```

```
[1] "df"       "df_clean" "raw_data"
```